R13

Code No:111AL

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year Examinations, December - 2018
MATHEMATICAL METHODS
(Common to EEE, ECE, CSE, EIE, IT)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

Part- A

(25 Marks)

1.a) Prove that $E=e^{hD}$ where D is a differential operator. [2]

b) Write the Normal equations for fitting a curve $y = a + \frac{b}{x}$ from the given data points (x_i, y_i). [3]

c) Explain the solution of the equation $x+1=0$ graphically. [2]

d) Evaluate $\int_0^1 \frac{dx}{1+x}$ using Trapezoidal Rule. [3]

e) Write the finite Fourier cosine transform of f(x) in $[0,\pi]$. [2]

f) Find the half range sine series of f(x) = x in [0,1]. [3]

g) Form PDE by eliminating constants from z=ax+by+ab . [2]

h) Solve the PDE px+qy = 1. [3]

i) State stokes theorem. [2]

j) Find the angle between the surface $x^2+y^2+z^2=9$ and $z=x^2+y^2-3$ at the point (2,-1,2). [3]

Part - B

(50 Marks)

2.a) Find f(2.5) from the following table using Gauss Forward interpolation formula.

X	1	2	3	4
Y	11	20	44	79

b) Fit the curve $y = a + bx + cx^2$ from the following table [5+5]

X	1	2	3	4
Y	0	1	2	3

OR

3.a) Find f(0.5) from the following table using Newton forward interpolation formula.

X	0	1	2	3
Y	7	10	15	34

b) Find polynomial which is passing through the following points
(1, 21),(3,15),(4,18),(6,25)

[5+5]

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4.a) Find a positive root of $2^x - 3x = 0$ using Bisection Method.

b) Solve the system of equation using LU decomposition method (Crout's)
x+5y+z = 14, 2x+y+3z = 13, 3x+y+4z = 17. [5+5]

OR

5.a) Evaluate y (1.2) and y (1.4) by RK method of fourth order if $y' = 1 + xy, y(1) = 1$.

b) Solve the system of equation using Gauss –seidel method:
8x-3y+2z = 20, 4x+11y-z = 33, 6x+3y+12z = 35. [5+5]

6.a) Find the Fourier series of $f(x) = \begin{cases} 0, & -\pi < x < 0 \\ 1, & 0 < x < \pi \end{cases}$

Hence deduce that $1 - \frac{1}{3} + \frac{1}{5} + \ldots = \frac{\pi}{4}$

b) Find the Fourier sine transform of $f(x)$ defined by $f(x) = \begin{cases} x & 0 < x < 1 \\ 1 - x & 1 < x < 2 \\ 0, & otherwise \end{cases}$ [5+5]

OR

7.a) Using Fourier integral, Show that $\int_0^\infty \frac{\sin \pi\lambda}{1-\lambda^2} \sin \lambda x \, d\lambda = \begin{cases} \frac{1}{2}\pi \sin x & if \ 0 < x < \pi \\ 0 & if \ x > \pi \end{cases}$

b) Find the Half range cosine series for f(x) = $x+x^2$ in [0, 2] [5+5]

8.a) Solve the PDE $z\left(x^2 - y^2\right) = px^2 - qy^2$

b) Solve $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u, u(x,0) = 6e^{-3x}$ by method of separation of variables [5+5]

OR

9.a) Solve the PDE $p^2 y(1+x^2) = qx^2$

b) Solve the Laplace equation $u_{xx} + u_{yy} = 0$ with the following boundary conditions
$u(0, y) = 0 \quad for \ 0 \le y \le \pi$
$u(\pi, y) = 0 \quad for \ 0 \le y \le \pi$
$u(x, \pi) = f(x) \quad for \ 0 \le x \le \pi$
$u(x, 0) = 0 \quad for \ 0 \le x \le \pi$ [5+5]

10.a) If f and g are any two vector functions, then show that
$\nabla \times (f \times g) = (g \cdot \nabla) \ f - (f \cdot \nabla) g + f div g - g div f$

b) Evaluate $\int_c (xy - y^2)dx + x^2 y dy$ along the closed curve formed by y = 0 , x =1 and y = x by Green's theorem. [5+5]

OR

11.a) Prove that div curl $\bar{f} = 0$ where f is a vector point function.

b) Evaluate $\iint_S \bar{A}.\bar{n} ds$ where $\bar{A} = z\bar{i} + x\bar{j} - 3y^2 z\bar{k}$ and S is the surface of the cylinder $x^2+y^2=16$ included in the first octant between z=0 and z=5. [5+5]

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